

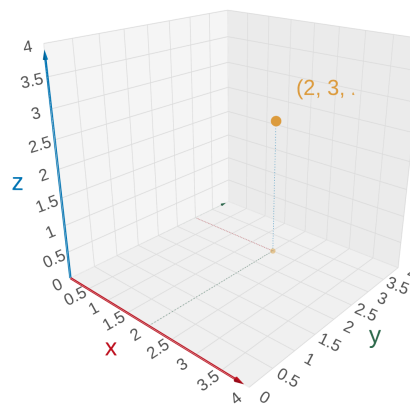
Section 12.1: 3D Coordinate Systems

Vectors and the Geometry of Space

MTH 310
Calculus III

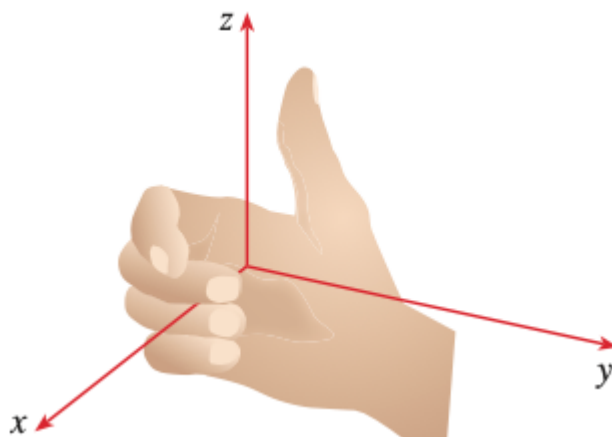
Ordered Triples in 3D

We use a **right-handed** coordinate system.



[Interactive version](#)

The Right-Hand Rule

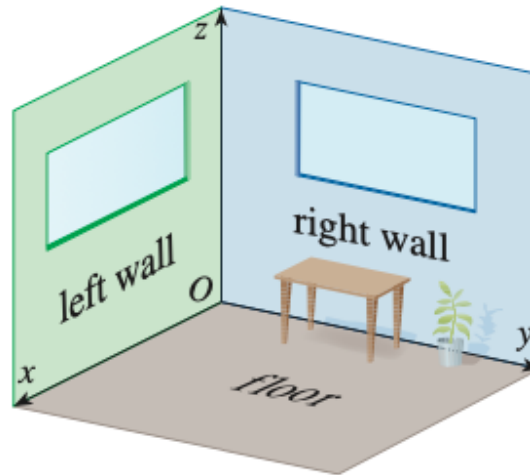


Classroom Coordinates

Example 1

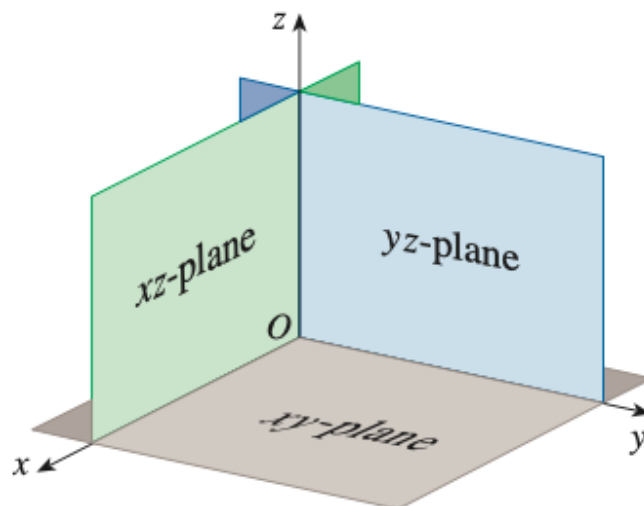
If the lower-left corner at the front of the classroom is the **origin**, with coordinate axes along the edges formed by the floor and walls, what are the approximate coordinates (in feet) of:

- The projector?
- The nearest door in Murphy Library?

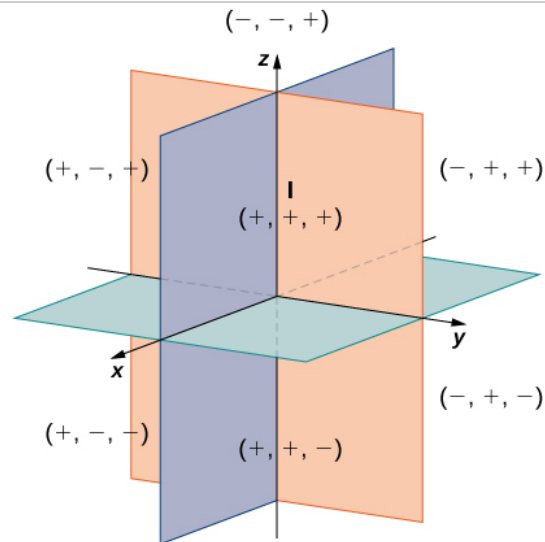


Estimate the coordinates of each object.

Coordinate Planes

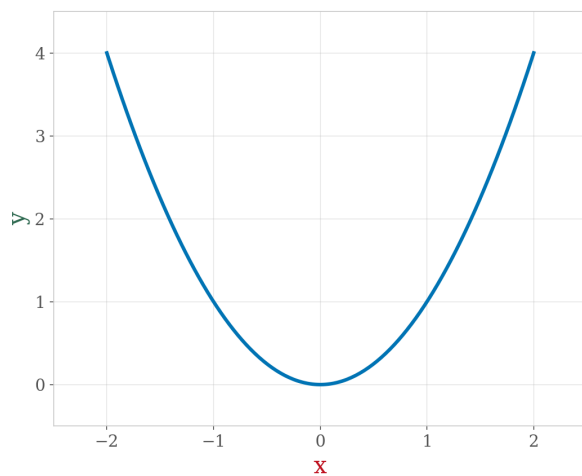


The Eight Octants

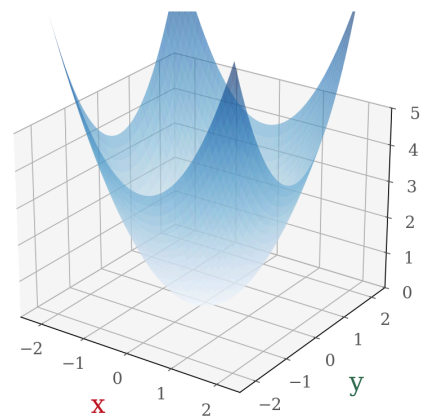


Curves vs. Surfaces

In **2D**: an equation in x, y describes a **curve** in $\mathbb{R}^2$
In **3D**: an equation in x, y, z describes a **surface** in $\mathbb{R}^3$



$y = x^2$ — a curve in $\mathbb{R}^2$



$z = x^2 + y^2$ — a surface in $\mathbb{R}^3$

Sketching Planes

Example 2

Identify and sketch the surfaces $z = 3$ and $y = 5$.

Sketch each plane in a 3D coordinate system.

A Cylinder

Example 3

Identify and sketch the surface $x^2 + z^2 = 4$.

What does this equation look like in 2D? What happens in 3D?

Bonus: Parabolic Cylinder

Sketch $z = x$ -squared in the xz -plane, then think about what happens in 3D.

Key Idea: Cylinders

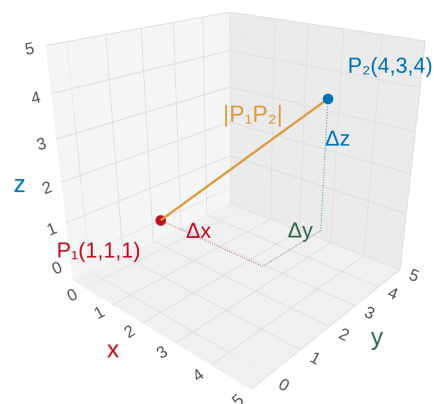
When an equation is **missing one variable** in $\mathbb{R}^3$, the graph is a **cylinder** — the 2D curve extended along the missing axis.

Distance Formula in 3D

Distance Formula in Three Dimensions

The distance $|P_1P_2|$ between points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is:

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$



[Interactive version](#)

Equation of a Sphere

Example 4

What is the equation for a sphere of radius r centered at $C(h, k, l)$?

Use the distance formula with $|PC| = r$. Square both sides.

Describing a 3D Region

Example 5

Describe and sketch the region in $\mathbb{R}^3$ described by:

$$x^2 + z^2 \leq 25, \quad 0 \leq y \leq 2$$

Start with the 2D circle in the xz-plane. Then think about what the cylinder looks like. Finally, apply the y-bounds.

Homework

Section 12.1: 2, 9, 11, 13a, 17, 19, 31, 35, 39, 41, 43, 45